9th Class • Punjab Board • PECTAA New Syllabus

Chapter 9: Entrepreneurship in Digital Age

Complete solved exercise — digital tools, design thinking, business planning, ethics and SDGs.

Chapter 9 • Complete official exercise • 8 MCQs + 7 short + 5 long questions

What this chapter is about

• **Entrepreneurship** turns an identified need or opportunity into an organised venture. Entrepreneurs combine innovation, initiative, calculated risk-taking and persistence.

• **Digital entrepreneurship** uses websites, apps, social media, online marketing, data analytics and e-commerce to reach and serve customers.

• **Design thinking** develops solutions through empathy, problem definition, ideation, prototyping and testing. A business plan then connects market analysis, revenue and marketing with operations and finance.

• Ethical and sustainable entrepreneurship balances profit with honesty, fairness, employee/community welfare and environmental responsibility, including relevant Sustainable Development Goals (SDGs).

Coverage: Chapter 9 corresponds to Unit 12 Entrepreneurship in Digital Age in the twelve-unit textbook. The April 2026 nine-unit exercise is solved in book order.

A. Multiple Choice Questions

1. What is entrepreneurship?

(a) The process of starting a new business or organization
(b) The process of buying and selling stocks
(c) The process of working for a large company
(d) The process of creating a marketing campaign

Answer: (a) The process of starting a new business or organization — Entrepreneurship involves identifying an opportunity and organising resources to create and run a new venture or organisation.

2. What is a key characteristic of entrepreneurs?

(a) Avoiding risks
(b) Seeking job security
(c) Innovation
(d) Following established methods

Answer: (c) Innovation — Entrepreneurs create or improve products, services and methods; they manage calculated risk rather than simply avoiding it.

3. An example of a digital technology used by modern entrepreneurs is:

(a) Newspaper advertisements
(b) Television commercials
(c) Mobile apps
(d) Door-to-door sales

Answer: (c) Mobile apps — Mobile apps let ventures sell products, deliver services, communicate and collect useful feedback digitally.

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4. Which of the following is NOT a technique for identifying market needs?

(a) Surveys and Questionnaires
(b) Market Research
(c) Observation
(d) Brainstorming

Answer: (d) Brainstorming — In the textbook's list, surveys, research and observation directly investigate needs; brainstorming is primarily idea generation, not direct market-need evidence.

5. The first step in the design thinking process is:

(a) Prototyping
(b) Testing
(c) Empathizing
(d) Defining

Answer: (c) Empathizing — Design thinking begins by understanding users' experiences and needs before defining the problem.

6. The primary purpose of creating a business plan is:

(a) To attract investors
(b) To outline strategies for growth
(c) To study market trends
(d) To implement digital marketing strategies

Answer: (b) To outline strategies for growth — A business plan is a roadmap that sets goals and strategies for operation and growth; attracting investors is only one possible use.

7. What does market analysis involve?

(a) Calculating revenue forecasts
(b) Researching competitors and understanding customer needs
(c) Developing pricing strategies
(d) Implementing digital marketing campaigns

Answer: (b) Researching competitors and understanding customer needs — Market analysis studies customers, demand, trends and competitors so the business can make informed choices.

8. Which Sustainable Development Goal (SDG) focuses on affordable and clean energy?

(a) SDG5
(b) SDG7
(c) SDG12
(d) SDG17

Answer: (b) SDG7 — SDG 7 is Affordable and Clean Energy.

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B. Short Questions

1. What is the meaning of the word entrepreneur?

An entrepreneur is a person who identifies an opportunity and starts or develops a business or organisation, organising resources and accepting calculated risk. The word comes from the French *entreprendre*, meaning “to undertake.”

2. How do digital technologies help entrepreneurs reach a global audience?

Websites, apps, social media, search engines and online marketplaces make a business visible beyond its local area. Digital advertising can target suitable audiences, while online communication, payment and delivery tools help serve customers in different countries at relatively low cost.

3. What is an e-commerce platform?

An e-commerce platform is an online system that allows businesses and customers to list, search, order and pay for products or services. It may also manage a catalogue, shopping cart, payments, orders, delivery status and customer support; examples include a business's own store or an online marketplace.

4. Why is it important to identify market needs when generating a business idea?

A business succeeds when it solves a real problem or meets a genuine demand. Identifying needs reduces the risk of making an unwanted product, reveals customer preferences and gaps in existing offers, and guides useful features, pricing and marketing.

5. What is the main purpose of using design thinking in creative problem-solving?

Design thinking creates solutions around real user needs. Teams empathise with users, define the actual problem, generate ideas, build simple prototypes and test them, using feedback to improve the solution before investing heavily.

6. Write the importance of market analysis in a business plan.

Market analysis shows who the customers are, what they need, how large demand may be and which competitors already serve them. This evidence guides product design, pricing, promotion and sales forecasts and helps identify risks and a realistic competitive advantage.

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7. How can businesses contribute to environmental sustainability according to the Sustainable Development Goals?

Businesses can reduce energy and water use, choose renewable energy, minimise waste, reuse and recycle materials, design durable products and measure their environmental impact. These actions support goals such as SDG 7 (clean energy), SDG 12 (responsible consumption and production) and SDG 13 (climate action).

C. Long Questions

1. Explain why entrepreneurship is important for economic growth and innovation.

Entrepreneurship converts ideas and underused resources into productive goods and services. A well-managed venture creates value for customers as well as income for its owners.

• **Job creation:** New and expanding businesses hire employees and can support additional work among suppliers, transporters and service providers.

• **Innovation:** Entrepreneurs develop new products or improve existing processes to solve unmet needs, often using digital technology in creative ways.

• **Productivity:** Better tools, business models and organisation allow more output from the same time, labour or materials.

• **Competition and consumer choice:** New ventures challenge established firms to improve quality, service and prices while giving customers more options.

• **Investment and income:** Successful ventures attract capital, pay wages and taxes and create income that circulates through the wider economy.

• **Regional and social development:** Local and social enterprises can bring services, skills and opportunities to underserved communities.

• **Digital markets and exports:** Online platforms let even small firms reach national and international customers, supporting diversification and potential foreign earnings.

For example, a small Pakistani startup that builds a low-cost irrigation advisory app can sell a useful service, employ developers and support farmers in reducing wasted water. Entrepreneurship involves risk and not every venture grows, so planning, ethics and evidence remain essential.

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2. Explain how market research tools, online marketing tools, and e-commerce platforms can work together to help a business succeed in the digital world.

These tools form a connected cycle: **learn about customers → reach them with a suitable offer → sell and serve online → measure and improve**.

Figure 1 — Research, marketing and sales inform one another



Tool group	Role in the cycle	Examples
Market research tools	Collect customer needs, competitor information, search trends and feedback.	Surveys/questionnaires, interviews, Google Forms, web analytics and social listening.
Online marketing tools	Use research to prepare relevant content and reach selected audiences, then measure response.	SEO tools, social media, email marketing, content calendars and advertising dashboards.
E-commerce platforms	Turn interest into convenient browsing, ordering, payment, delivery tracking and support.	Business website/store, mobile app and online marketplace.
Analytics and customer feedback	Measure visits, conversion, repeat purchases, returns and satisfaction to guide the next decision.	Store analytics, reviews and post-purchase surveys.

Example: a small online bookstore

A survey may show that students want affordable bundles of Class 9 books. The business uses search-friendly pages and targeted social posts to promote the bundles, sells them through its e-commerce store and offers online payment plus delivery tracking. Store data may show that many visitors leave at checkout; customer feedback can reveal delivery-cost concerns, leading to a revised offer. Research therefore improves marketing, marketing brings suitable visitors, the platform enables sales and sales data guide further research.

3. Choose a local business in Pakistan and discuss how it aligns with one or more Sustainable Development Goals (SDGs).

Model answer: “Lahore Solar Solutions” is a fictional local solar-installation and energy-audit business created for this response, not a claim about a real company.

Assume that Lahore Solar Solutions installs appropriately sized rooftop solar systems, conducts energy audits and trains local technicians. Its SDG contribution can be evaluated through its environmental, social and economic effects.

SDG alignment	Business action	Expected impact and evidence
SDG 7 — Affordable and Clean Energy	Install safe solar systems and recommend efficient appliances after an energy audit.	Track renewable electricity generated and reduced dependence on high-carbon grid electricity; make efficiency guidance accessible.
SDG 8 — Decent Work and Economic Growth	Train and fairly employ local technicians and buy suitable supplies through responsible local firms.	Record skilled jobs, training hours, safe-working compliance and fair wages.
SDG 12 — Responsible Consumption and Production	Size systems correctly, maintain equipment, reduce packaging and use authorised recycling for batteries and electronic parts.	Measure waste diverted, maintenance completed and equipment life; avoid exaggerated green claims.
SDG 13 — Climate Action	Help customers understand and reduce energy-related emissions without ignoring panel/battery disposal.	Estimate avoided emissions using transparent assumptions and maintain an end-of-life disposal plan.

• **Environmental:** Greater renewable-energy use and efficiency can reduce emissions and resource waste, while responsible recycling limits a new waste problem.

• **Social:** Technician training, safe work and reliable power for suitable customers can improve skills and daily life; financing options must be transparent and fair.

• **Economic:** Customers may lower long-term energy costs, the business earns revenue and local workers/suppliers gain income.

Alignment should be measured rather than assumed. The business must report both benefits and negative effects, such as battery waste or unsafe installation, and take corrective action.

4. Explain digital entrepreneurship, also write the principles of ethics in Business.

Digital entrepreneurship is the creation and operation of a venture that uses digital technologies as an important part of its product, service, marketing, sales or management. Examples include an online bookstore, a mobile tutoring service and a local shop selling nationwide through an app.

• Digital tools reduce the cost of reaching customers, support remote work and enable rapid feedback and data-based decisions.

• Common tools include websites, mobile apps, e-commerce platforms, cloud documents, digital payments, analytics, SEO, email and social-media marketing.

• Success still requires a genuine need, a workable revenue model, reliable service and customer trust; technology alone is not a business model.

Ethical principle	How a digital business applies it
Honesty	Describe products, prices, delivery times, reviews and expected results truthfully; avoid fake claims and hidden charges.
Integrity	Keep promises and apply the same ethical standards even when a shortcut would increase profit or remain unseen.
Fairness	Provide equal and reasonable treatment, accessible opportunities and transparent terms to customers, workers and suppliers.
Respect	Value people's dignity, rights, time, culture and property, including copyright and safe workplace behaviour.
Privacy and confidentiality	Collect minimum personal data, explain its use, protect it and never sell or disclose it improperly.
Responsibility and sustainability	Consider employees, community and the environment; correct mistakes, handle complaints and reduce waste.

5. Write down the components of a Business plane.

A business plan is a structured roadmap that explains the venture's goals and how it will achieve them. Its exact length depends on the business, but a comprehensive plan normally contains the following components.

Component	What it explains
Executive summary	A short overview of the idea, customer problem, product, target market, advantage, main financial need and goals.
Business description and objectives	Mission, legal/organisational form, location, vision and measurable short-/long-term objectives.
Products or services	Features, benefits, development stage, pricing logic, intellectual-property needs and how the offer solves the customer problem.
Market analysis	Target customers, needs, market size/trends, competitors and evidence from surveys, interviews or research.
Marketing and sales strategy	Positioning, promotion channels, digital marketing, sales process, customer acquisition and retention.
Operations plan	Location, suppliers, equipment, technology, staffing, production/delivery process, quality and customer support.

Component	What it explains
Management and organisation	Founders' roles, required skills, staff structure, responsibilities and advisers.
Revenue model and financial plan	Income sources, prices, costs, sales forecast, cash flow, break-even estimate, startup funding and profit/loss expectations.
Risk, ethics and sustainability	Major market, operational, financial, security and environmental risks, with mitigation and responsible practices.
Milestones and appendix	Timeline, prototype/testing targets, launch and review dates, plus supporting research, designs or detailed forecasts.

Prototyping and iteration

Before full investment, build a simple prototype or minimum version of the offer and show it to potential customers. Use feedback to improve the product and revise the plan. A business plan is therefore a living document, not a one-time school exercise.

D. More Important Questions (from the text)

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1. Explain the design thinking process with a rural Pakistan example.

Figure 2 — Design thinking is iterative



1. Empathise: Listen to farmers and observe their work to understand water availability, literacy, phone access, costs and timing problems.

2. Define: State a user-centred problem, such as “Small farmers need timely, understandable irrigation guidance on basic phones.”

3. Ideate: Generate several possible answers—SMS advice, a voice helpline, local-language audio messages or a community weather board—before choosing one.

4. Prototype: Create a cheap sample, such as one week of mock SMS/audio irrigation messages, without developing a complete app.

5. Test: Ask farmers to use the messages, record confusion and results, then revise wording, timing or channel and repeat the cycle.

The process reduces the risk of building an advanced app that farmers cannot or do not want to use.

2. Compare common digital revenue models.

Revenue model	How income is earned	Example
Direct sales	Customers pay for each product, service or order.	Selling a book through an online store.
Subscription	Customers pay regularly for continuing access or service.	Monthly access to an online revision library.
Freemium	A basic service is free; advanced features require payment.	Free notes with a paid ad-free practice pack.
Advertising / sponsorship	Advertisers or sponsors pay for suitable exposure to the audience.	Sponsored educational content, clearly disclosed.
Commission / marketplace fee	The platform takes a fee from transactions between buyers and sellers.	A marketplace charging a percentage per order.
Service / licensing fees	Customers pay for expertise, custom work or permission to use intellectual property.	Website setup or licensing original software.

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Choose a model that matches customer value and costs. A free service with many users is not automatically profitable, and hidden charges damage trust.

3. Official-unit extension: Outline a business plan for a small online bookstore for students.

Required part	Model plan
Market analysis	Survey local students about required books, price range and delivery preferences. Compare local shops and online sellers on price, availability, delivery time and return policy. Target Class 9–10 students who want convenient, affordable bundles.
Revenue model	Earn margin on new books and carefully checked used books; offer discounted subject bundles. Estimate costs for purchasing stock, packaging, delivery, website, marketing and returns before setting prices and cash-flow forecasts.
Digital marketing	Use search-friendly book pages, useful book lists, school-community social posts, permission-based email/WhatsApp updates and honest reviews. Measure visits, orders and repeat purchases rather than likes alone.
Operations and ethics	Maintain accurate stock records, secure payments and personal data, confirm delivery charges, provide support/returns and package orders with minimal recyclable material.
Prototype and review	Begin with a small Class 9 catalogue and local delivery area. Test a simple store page, gather customer feedback and expand only after checking demand, profit and service reliability.

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Points to Remember

Entrepreneur: a person who undertakes and organises a venture, accepting calculated risk; innovation is a key characteristic.

Digital entrepreneurship: uses apps, websites, e-commerce, digital marketing and analytics to create and run a venture.

Market analysis: researches customer needs, demand, trends and competitors before major decisions.

Design thinking: empathise → define → ideate → prototype → test → improve; begin with users, not technology.

Business plan: links objectives, product, market, marketing, operations, team, revenue/finance, risks and milestones.

Revenue: direct sales, subscriptions, freemium, advertising, commission and service/licensing are different models.

Ethics: honesty, integrity, fairness, respect, privacy and responsibility build customer trust.

SDG 7: Affordable and Clean Energy; sustainable firms also consider SDG 8, 12 and 13 where relevant.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	a	2	c	3	c	4	d	5	c
6	b	7	b	8	b				

How to attempt this chapter in the paper

• For a business-plan answer, include numbers where possible: target customers, price, main costs, funding, milestones and measures of success.
• For design thinking, do not begin with ideation; empathy and a clear user-centred problem definition come first.
• For an SDG answer, connect specific business actions to social, environmental and economic effects and explain how impact will be measured.
• For digital-tools answers, show the complete feedback cycle from research to marketing, sales, analytics and improvement.

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